

# MGLC-XA Architecture Review Checklist

Design Validation, Performance Verification, Integration & Risk Assessment

**System:** Multi-Layer Graph-Logic Controller — eXtended Acceleration (MGLC-XA)

**Document Type:** Architecture Review Checklist

**Version:** 1.0 | **Date:** August 2026

**Applicable Gates:** Gate 1 (Design Freeze) | Gate 2 (FPGA Validation) | Gate 3 (ASIC Tape-Out) | Gate 4 (Production Release)

 **CLASSIFICATION: INTERNAL TECHNICAL USE ONLY**

Not for distribution outside authorized review personnel. Handle in accordance with internal information classification policy.

## Revision History

| Version | Date        | Author         | Change Description                                                                 |
|---------|-------------|----------------|------------------------------------------------------------------------------------|
| 1.0     | August 2026 | Emanuel Schaaf | Initial release — baseline architecture review checklist for all four gate stages. |
| 0.9     | July 2026   | Emanuel Schaaf | Draft for internal peer review. Risk assessment section                            |

| Version | Date       | Author         | Change Description                                                            |
|---------|------------|----------------|-------------------------------------------------------------------------------|
|         |            |                | expanded.                                                                     |
| 0.8     | June 2026  | Emanuel Schaaf | Integration section added. Security risk subsection restructured.             |
| 0.5     | April 2026 | Emanuel Schaaf | Initial draft — Design Architecture Validation and Performance sections only. |

## HOW TO USE THIS DOCUMENT

### Purpose Statement

This checklist is the primary instrument used to systematically validate that the **MGLC-XA (Multi-Layer Graph-Logic Controller — eXtended Acceleration)** architecture meets all design, performance, integration, safety, and risk requirements prior to milestone gate decisions. It is intended for use by internal architecture review boards, external technical auditors, system integrators, and design validation teams.

Each gate review requires a completed, signed instance of this document. Items marked *Fail* or *Partial* at a given gate must have documented resolution paths with assigned owners and target dates before a gate decision can be rendered.

### Gate Stages Covered

| Gate          | Name            | Trigger Condition                                       | Primary Sections Required                  |
|---------------|-----------------|---------------------------------------------------------|--------------------------------------------|
| <b>Gate 1</b> | Design Freeze   | All architectural specifications complete and baselined | Section 1, Section 4                       |
| <b>Gate 2</b> | FPGA Validation | FPGA prototype functional, performance                  | Section 1, Section 2, Section 3, Section 4 |

| Gate          | Name               | Trigger Condition                                           | Primary Sections Required |
|---------------|--------------------|-------------------------------------------------------------|---------------------------|
|               |                    | benchmarks executed                                         |                           |
| <b>Gate 3</b> | ASIC Tape-Out      | RTL freeze, DFT complete, sign-off from all technical leads | All Sections              |
| <b>Gate 4</b> | Production Release | Silicon characterization complete, software stack validated | All Sections              |

### Status Legend

| Symbol                                                                              | Status         | Definition                                                                    |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------|
|    | <b>Pass</b>    | Requirement fully met. Supporting evidence referenced in Notes field.         |
|   | <b>Fail</b>    | Requirement not met. Blocks gate progression. Resolution plan mandatory.      |
|  | <b>Partial</b> | Requirement partially met. Conditions and open gaps must be documented.       |
|  | N/A            | Not applicable to this product variant or gate stage. Justification required. |
|  | <b>Pending</b> | Review not yet completed. Not acceptable for gate sign-off.                   |

### Instructions for Reviewers

#### Reviewer Instructions

1. Each reviewer is assigned specific sections per their domain of expertise (see Section 5: Sign-Off).
2. For every checklist item, record a Status value using the legend above, an

Owner name or team, and supporting evidence or notes in the Notes/Evidence column.

3. Items left as **Pending** at gate review time constitute a gate blocker. All Pending items must be resolved or formally deferred via the Open Issues Log before the gate decision is signed.

4. Do not alter the Item # or Checklist Item description. Add supplemental notes in the Notes/Evidence column only.

5. Completed checklists must be submitted to the Review Board Chair no fewer than **5 business days** prior to the scheduled gate review meeting.

6. All Fail and Partial items require an entry in the Open Issues Log (Section 5.2) with assigned severity, owner, and target resolution date.

## Checklist Summary Scorecard

Complete this scorecard after all checklist items have been reviewed. Score % = Pass / (Total Items – N/A) × 100.

| Section                                                | Total Items | ✓ Pass | ✗ Fail | ⚠ Partial | □ Pending | — N/A | Score % |
|--------------------------------------------------------|-------------|--------|--------|-----------|-----------|-------|---------|
| <b>Section 1:<br/>Design Architecture Validation</b>   | 50          |        |        |           |           |       |         |
| <b>Section 2:<br/>Performance Targets Verification</b> | 26          |        |        |           |           |       |         |
| <b>Section 3:<br/>Integra</b>                          | 30          |        |        |           |           |       |         |

| Section                    | Total Items |  Pass |  Fail |  Partial |  Pending |  N/A | Score % |
|----------------------------|-------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------|
| tion Requirements          |             |                                                                                        |                                                                                        |                                                                                           |                                                                                            |                                                                                         |         |
| Section 4: Risk Assessment | 33          |                                                                                        |                                                                                        |                                                                                           |                                                                                            |                                                                                         |         |
| TOTAL                      | 139         |                                                                                        |                                                                                        |                                                                                           |                                                                                            |                                                                                         |         |

SECTION 1: DESIGN ARCHITECTURE VALIDATION

This section validates the completeness, correctness, and formal specification of the MGLC-XA architecture across all three primary layers and the memory subsystem. All items in this section are required for Gate 1 (Design Freeze) and subsequent gates.

### 1.1 Architectural Specification Completeness

| #     | Checklist Item                                                                                                                                   | Status                                                                                      | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------|------------------|
| 1.1.1 | Full architectural specification document exists, is formally released, and is under version control in the approved document management system. |  Pending |       |                  |
| 1.1.2 | All three primary layers are formally defined in the specification: Multi-Layer                                                                  |  Pending |       |                  |

| #            | Checklist Item                                                                                                                                                           | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|              | Graph Engine (MLGE), Logic Control Matrix (LCM), and eXtended Acceleration Units (XAU).                                                                                  |                                  |       |                  |
| <b>1.1.3</b> | All inter-layer interfaces are formally specified, including protocol type, maximum bandwidth, and latency bounds for each interface pair (MLGE↔LCM, LCM↔XAU, MLGE↔XAU). | <input type="checkbox"/> Pending |       |                  |
| <b>1.1.4</b> | Memory model is coherent across all three layers; shared memory regions, ownership, and access permissions are unambiguously defined.                                    | <input type="checkbox"/> Pending |       |                  |
| <b>1.1.5</b> | Addressing scheme is fully defined and verified to be collision-free across all layers and memory regions.                                                               | <input type="checkbox"/> Pending |       |                  |
| <b>1.1.6</b> | State machine diagrams exist for all control flows within MLGE, LCM, and XAU; diagrams are version-controlled and linked from the architectural specification.           | <input type="checkbox"/> Pending |       |                  |

| #      | Checklist Item                                                                                                                                                                | Status                           | Owner | Notes / Evidence |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 1.1.7  | Exception and error handling paths are fully specified for all error conditions in all three layers, including propagation, isolation, and recovery behavior.                 | <input type="checkbox"/> Pending |       |                  |
| 1.1.8  | Boot sequence and initialization order is documented, including dependency ordering between layers, power domain sequencing, and safe state definitions.                      | <input type="checkbox"/> Pending |       |                  |
| 1.1.9  | Power domain mapping is complete; all voltage rails, power domains, and domain boundaries are identified and cross-referenced with the physical design plan.                  | <input type="checkbox"/> Pending |       |                  |
| 1.1.10 | Clock Domain Crossing (CDC) analysis is complete; all asynchronous boundaries are identified, synchronization strategies are documented, and formal CDC sign-off is recorded. | <input type="checkbox"/> Pending |       |                  |

## 1.2 MLGE (Multi-Layer Graph Engine) Validation

| #     | Checklist Item                                                                                                                                                                                         | Status                           | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 1.2.1 | Graph traversal algorithm is formally specified, including supported modes: Breadth-First Search (BFS), Depth-First Search (DFS), and any custom walk modes; algorithmic complexity bounds are stated. | <input type="checkbox"/> Pending |       |                  |
| 1.2.2 | Both sparse and dense adjacency representation are supported and validated; switching between representations is correct and incurs no data loss.                                                      | <input type="checkbox"/> Pending |       |                  |
| 1.2.3 | Graph memory layout formats validated for correctness and performance: Compressed Sparse Row (CSR), Compressed Sparse Column (CSC), and Coordinate (COO) formats all supported per specification.      | <input type="checkbox"/> Pending |       |                  |
| 1.2.4 | Maximum graph size bounds are defined (vertex count, edge                                                                                                                                              | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                    | Status                           | Owner | Notes / Evidence |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | count, attribute data size); hardware enforcement of bounds is implemented and tested.                                                                                                            |                                  |       |                  |
| 1.2.5 | Graph traversal throughput target of $\geq 2.0$ TB/s effective memory bandwidth is validated by RTL simulation or FPGA emulation; results are recorded with test configuration details.           | <input type="checkbox"/> Pending |       |                  |
| 1.2.6 | Edge weight and label handling is supported per specification; correctness verified through simulation with representative workloads including signed, unsigned, and floating-point edge weights. | <input type="checkbox"/> Pending |       |                  |
| 1.2.7 | Multi-hop traversal depth limit is defined in the specification; hardware enforcement is implemented; behavior at depth limit (truncation, exception, or flag) is specified and tested.           | <input type="checkbox"/> Pending |       |                  |
| 1.2.8 | Graph                                                                                                                                                                                             | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | partitioning logic is correct for distributed subgraph topologies; partition boundaries, cross-partition edge handling, and consistency guarantees are verified by simulation.        |                                  |       |                  |
| <b>1.2.9</b>  | Deadlock-freedom of the traversal engine is formally verified using model checking or exhaustive simulation; all identified potential deadlock scenarios are documented and resolved. | <input type="checkbox"/> Pending |       |                  |
| <b>1.2.10</b> | Latency bounds per traversal operation are specified (best-case, worst-case, average-case); bounds are validated by simulation under representative workloads including stress cases. | <input type="checkbox"/> Pending |       |                  |

### 1.3 LCM (Logic Control Matrix) Validation

| #            | Checklist Item                    | Status                           | Owner | Notes / Evidence |
|--------------|-----------------------------------|----------------------------------|-------|------------------|
| <b>1.3.1</b> | Reconfiguration protocol is fully | <input type="checkbox"/> Pending |       |                  |

| #            | Checklist Item                                                                                                                                                                                   | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|              | specified, including initiation sequence, acknowledgment handshake, timeout behavior, and abort/rollback procedures.                                                                             |                                  |       |                  |
| <b>1.3.2</b> | Reconfiguration latency target of <100 $\mu$ s is validated by RTL simulation or FPGA emulation; results recorded under both typical and worst-case configuration payload sizes.                 | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.3</b> | Logic plane isolation during reconfiguration is verified: no data corruption, no unintended signal assertion, and no bus contention occurs on active paths while reconfiguration is in progress. | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.4</b> | All supported dataflow path combinations through the LCM are enumerated, tested, and documented; no untested combinations exist in the supported configuration space.                            | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.5</b> | Configuration state backup                                                                                                                                                                       | <input type="checkbox"/> Pending |       |                  |

| #            | Checklist Item                                                                                                                                                           | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|              | and restore are implemented and tested; restored configuration is bit-for-bit identical to the saved state; test vectors and pass criteria are documented.               |                                  |       |                  |
| <b>1.3.6</b> | Power-on default configuration is defined, documented as a safe operational state, and validated to not assert any unintended signals upon device power-up.              | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.7</b> | LCM arbitration logic is formally verified to be deadlock-free under all combinations of simultaneous reconfiguration requests from competing requestors.                | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.8</b> | Configuration memory integrity checking via ECC or parity is implemented, covers all configuration memory arrays, and has been verified through fault injection testing. | <input type="checkbox"/> Pending |       |                  |
| <b>1.3.9</b> | Partial reconfiguration support is                                                                                                                                       | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                | Status                           | Owner | Notes / Evidence |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | validated: partial updates apply correctly without disturbing non-targeted configuration regions; correctness verified by targeted test suite.                                |                                  |       |                  |
| <b>1.3.10</b> | LCM-to-MLGE and LCM-to-XAU interface handshake protocols are verified for correctness under normal, back-pressure, and error conditions; verification results are documented. | <input type="checkbox"/> Pending |       |                  |

## 1.4 XAU (eXtended Acceleration Units) Validation

| #            | Checklist Item                                                                                                                                                                                  | Status                           | Owner | Notes / Evidence |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>1.4.1</b> | All domain-specific accelerators are enumerated in the specification: ML Inference XAU, Signal Processing XAU, and Control Loop XAU; no undocumented accelerator instances exist in the design. | <input type="checkbox"/> Pending |       |                  |
| <b>1.4.2</b> | Each XAU has a formal,                                                                                                                                                                          | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                               | Status                           | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | standalone functional specification document covering input/output interface, supported operations, precision modes, and boundary conditions.                                                                |                                  |       |                  |
| 1.4.3 | XAU Instruction Set Architecture (ISA) is documented for each programmable XAU variant; ISA includes opcode definitions, operand encoding, exception codes, and timing (latency/throughput) per instruction. | <input type="checkbox"/> Pending |       |                  |
| 1.4.4 | XAU compiler toolchain is verified against hardware behavior using co-simulation; compiled output for a representative set of kernels produces results bit-for-bit matching RTL simulation reference output. | <input type="checkbox"/> Pending |       |                  |
| 1.4.5 | ML Inference XAU throughput target of $\geq 3.0$ TOPS/W is validated by simulation; power and                                                                                                                | <input type="checkbox"/> Pending |       |                  |

| #            | Checklist Item                                                                                                                                                                                                      | Status                           | Owner | Notes / Evidence |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|              | throughput measurement methodology is documented; results reference the test workload specification.                                                                                                                |                                  |       |                  |
| <b>1.4.6</b> | XAU power gating is implemented and verified: power-gated XAU domains do not leak state, wake-up latency is within specification, and gating/ungating does not affect neighboring domains.                          | <input type="checkbox"/> Pending |       |                  |
| <b>1.4.7</b> | XAU error injection and fault detection mechanisms are implemented and tested; all injected faults are correctly detected and reported via the error signaling interface with no false negatives in the test suite. | <input type="checkbox"/> Pending |       |                  |
| <b>1.4.8</b> | XAU DMA engine correctness is verified: transfer completeness, address alignment, burst length handling, and error reporting are all validated through directed and random DMA                                      | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | test cases.                                                                                                                                                                                                 |                                  |       |                  |
| <b>1.4.9</b>  | XAU memory access conflicts between simultaneous DMA operations and processor-initiated accesses are resolved by arbitration; no data corruption or starvation is possible; verified by simulation.         | <input type="checkbox"/> Pending |       |                  |
| <b>1.4.10</b> | XAU firmware update mechanism is cryptographically authenticated, update integrity is verified prior to application, rollback to prior firmware is supported, and the mechanism has been tested end-to-end. | <input type="checkbox"/> Pending |       |                  |

## 1.5 Memory Subsystem

| #            | Checklist Item                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>1.5.1</b> | Unified coherent memory fabric specification is complete; fabric topology, arbitration policy, QoS configuration, and failover behavior are all documented. | <input type="checkbox"/> Pending |       |                  |
| <b>1.5.2</b> | Cache coherence                                                                                                                                             | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | protocol (MESI or equivalent) is fully specified and formally verified; coherence protocol transitions are tested under concurrent multi-initiator traffic patterns.                  |                                  |       |                  |
| 1.5.3 | Memory bandwidth allocation per layer is defined (minimum guaranteed bandwidth, peak burst bandwidth); allocation enforcement is implemented and verified under contention scenarios. | <input type="checkbox"/> Pending |       |                  |
| 1.5.4 | ECC implementation covers all on-chip SRAM arrays and all DRAM interfaces; ECC configuration (SECDED or stronger), detection/correction capability, and error reporting are verified. | <input type="checkbox"/> Pending |       |                  |
| 1.5.5 | Memory scrubbing interval is defined; scrubbing period is set to meet the specified reliability target;                                                                               | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                     | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | scrubbing impact on available memory bandwidth is characterized and within budget.                                                                                                                 |                                  |       |                  |
| 1.5.6 | DRAM refresh timing interference analysis is complete; worst-case refresh-induced latency spikes are quantified and either within specification or mitigated by design (refresh management logic). | <input type="checkbox"/> Pending |       |                  |
| 1.5.7 | Virtual-to-physical address translation latency is bounded; TLB miss penalty is specified and validated; IOMMU (if present) configuration and performance impact are documented.                   | <input type="checkbox"/> Pending |       |                  |
| 1.5.8 | Memory access security for multi-tenant deployments is verified: address space isolation between tenants is enforced at hardware level; isolation is verified by                                   | <input type="checkbox"/> Pending |       |                  |

| #      | Checklist Item                                                                                                                                                                               | Status                           | Owner | Notes / Evidence |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|        | penetration-style test cases.                                                                                                                                                                |                                  |       |                  |
| 1.5.9  | Out-of-bounds access detection is implemented at all memory interface points; out-of-bounds accesses generate an error interrupt and do not silently corrupt adjacent memory regions.        | <input type="checkbox"/> Pending |       |                  |
| 1.5.10 | Memory power state transitions (active, self-refresh, deep power-down) are verified; transition latencies are within specification; data retention through power state changes is validated. | <input type="checkbox"/> Pending |       |                  |

## SECTION 2: PERFORMANCE TARGETS VERIFICATION

This section verifies that the MGLC-XA meets all quantitative performance targets for throughput, latency, power, thermal behavior, and scalability. Simulation evidence and measurement methodology references are required for all Pass designations.

### Gate Requirement

All performance items marked as Pass for Gate 2 (FPGA Validation) or later must be supported by measured data from FPGA emulation or silicon characterization — not solely RTL simulation. Simulation-only evidence is acceptable only through Gate 1.

## 2.1 Throughput & Latency Benchmarks

| #     | Checklist Item                                                                                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 2.1.1 | Benchmark suite for MGLC-XA is formally defined, version-controlled, and approved by the architecture team; all performance claims in this section reference this benchmark suite.                                          | <input type="checkbox"/> Pending |       |                  |
| 2.1.2 | Graph traversal throughput $\geq 2.4$ TB/s effective memory bandwidth achieved under the standard graph benchmark workload; simulation result is recorded with test configuration, graph size, and traffic pattern details. | <input type="checkbox"/> Pending |       |                  |
| 2.1.3 | LCM reconfiguration latency $\leq 80 \mu\text{s}$                                                                                                                                                                           | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                         | Status                           | Owner | Notes / Evidence |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | measured from reconfiguration request assertion to reconfiguration-complete acknowledgment; result recorded for both minimum and maximum payload configurations.                                                       |                                  |       |                  |
| 2.1.4 | ML Inference XAU throughput $\geq 3.1$ TOPS/W measured under the defined ML inference benchmark; power measured at nominal voltage/frequency; result recorded with workload specification reference.                   | <input type="checkbox"/> Pending |       |                  |
| 2.1.5 | Control loop end-to-end latency $\leq 100 \mu\text{s}$ from input stimulus to actuator output command; verified through simulation with representative control loop workload; worst-case path identified and measured. | <input type="checkbox"/> Pending |       |                  |
| 2.1.6 | Memory read and write latency targets are defined per memory tier (L1 cache, L2 cache, on-chip SRAM, DRAM); measured                                                                                                   | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                             | Status                           | Owner | Notes / Evidence |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | latencies for each tier are within specification and documented.                                                                                                                                           |                                  |       |                  |
| 2.1.7 | Burst bandwidth can be sustained at peak rate for a minimum of 10 ms intervals without performance degradation; verified by sustained-load simulation with bandwidth monitoring throughout the interval.   | <input type="checkbox"/> Pending |       |                  |
| 2.1.8 | Pipeline stall rate is <5% under nominal load conditions; stall rate measured and recorded per pipeline stage; dominant stall sources identified and accepted or mitigated.                                | <input type="checkbox"/> Pending |       |                  |
| 2.1.9 | Back-pressure handling is verified at all inter-layer interfaces: no data loss, no deadlock, and no performance cliff occurs when a downstream layer applies sustained back-pressure to an upstream layer. | <input type="checkbox"/> Pending |       |                  |

## 2.2 Power & Thermal

| #     | Checklist Item                                                                                                                                                                                      | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 2.2.1 | Total power budget is defined and documented for all operating modes: idle/standby, nominal compute, and peak burst; each mode has a maximum power ceiling that is binding on the design.           | <input type="checkbox"/> Pending |       |                  |
| 2.2.2 | Peak system power $\leq$ target TDP (to be inserted per product variant) watts is verified by power simulation under the maximum-power workload vector; power simulation methodology is documented. | <input type="checkbox"/> Pending |       |                  |
| 2.2.3 | Per-domain power measurements are available for all power domains; domain-level power budget vs. actual comparison is reviewed and accepted by the power/thermal team.                              | <input type="checkbox"/> Pending |       |                  |
| 2.2.4 | Dynamic Voltage and Frequency Scaling (DVFS) is implemented                                                                                                                                         | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                       | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | and verified: all supported operating points are tested for stability; frequency/voltage transitions do not cause functional failures.                                                                               |                                  |       |                  |
| 2.2.5 | Thermal model is calibrated against simulation; model parameters (thermal resistance, junction-to-case, junction-to-board) are documented and reviewed by the thermal engineering team.                              | <input type="checkbox"/> Pending |       |                  |
| 2.2.6 | Maximum junction temperature target $T_j \leq 105^{\circ}\text{C}$ under peak power workload in the specified cooling configuration is verified by thermal simulation; simulation inputs and assumptions documented. | <input type="checkbox"/> Pending |       |                  |
| 2.2.7 | Thermal throttling thresholds and throttling behavior are defined; throttling is implemented and verified to activate                                                                                                | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                                    | Status                           | Owner | Notes / Evidence |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | correctly, reduce power within the required time, and restore performance when temperature drops.                                                                                                 |                                  |       |                  |
| <b>2.2.8</b>  | Power-on inrush current is within the board and system specification limits; inrush current profile is characterized by simulation and verified to not trip system-level protection circuits.     | <input type="checkbox"/> Pending |       |                  |
| <b>2.2.9</b>  | Leakage current is estimated for the target process node at the specified temperature corner; leakage contribution is within the total power budget and within process node specification limits. | <input type="checkbox"/> Pending |       |                  |
| <b>2.2.10</b> | Energy efficiency metric (TOPS/W or equivalent compute-efficiency metric) is validated against the stated competitive baseline; measurement methodology is consistent with industry standard      | <input type="checkbox"/> Pending |       |                  |

| # | Checklist Item          | Status | Owner | Notes / Evidence |
|---|-------------------------|--------|-------|------------------|
|   | benchmarking practices. |        |       |                  |

## 2.3 Scalability & Load Testing

| #            | Checklist Item                                                                                                                                                                                               | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>2.3.1</b> | System behavior is tested and characterized across the full range of graph sizes from minimum (single-vertex/edge) to maximum (at the hardware-enforced limit); no anomalous behavior at extreme boundaries. | <input type="checkbox"/> Pending |       |                  |
| <b>2.3.2</b> | Multi-instance (multi-chip or multi-partition) scaling behavior is characterized; throughput and latency scaling curves are documented from 1x to maximum supported instance count.                          | <input type="checkbox"/> Pending |       |                  |
| <b>2.3.3</b> | Worst-case latency under maximum sustained load is bounded and declared acceptable by the architecture review board; the bound is documented                                                                 | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | with the corresponding workload scenario.                                                                                                                                                             |                                  |       |                  |
| 2.3.4 | Performance degradation curve under oversubscription conditions is documented; degradation is graceful (no cliff edges) and performance recovery upon load reduction is within the specified time.    | <input type="checkbox"/> Pending |       |                  |
| 2.3.5 | Hot-path (critical timing path) identification is complete; all paths with slack <10% of the clock period are listed, reviewed, and either accepted or flagged for optimization.                      | <input type="checkbox"/> Pending |       |                  |
| 2.3.6 | No single-point-of-failure latency cliff edges are identified in the performance model; all queue-depth, buffer-fill, or arbitration-state transitions have gradual, predictable performance effects. | <input type="checkbox"/> Pending |       |                  |
| 2.3.7 | Performance behavior under thermal throttling conditions is documented;                                                                                                                               | <input type="checkbox"/> Pending |       |                  |

| # | Checklist Item                                                                                                                | Status | Owner | Notes / Evidence |
|---|-------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------|
|   | the performance envelope available at each throttling level is specified and acceptable for the target application use cases. |        |       |                  |

## SECTION 3: INTEGRATION REQUIREMENTS

This section validates that the MGLC-XA is fully specified and validated for integration into target hardware platforms, software stacks, and system environments. All items in this section are required at Gate 2 (FPGA Validation) and all subsequent gates.

### 3.1 Hardware Integration

| #            | Checklist Item                                                                                                                                                                 | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>3.1.1</b> | PCIe interface specification is complete, including generation (Gen 4 / Gen 5), lane count, BAR layout, power management capability (ASPM), and ACS/ARI support as applicable. | <input type="checkbox"/> Pending |       |                  |
| <b>3.1.2</b> | CXL interface support is defined (CXL 2.0 / CXL 3.0 as applicable), including memory                                                                                           | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | expander capability, device type classification, and coherency protocol support; or N/A is formally justified.                                                                                                        |                                  |       |                  |
| 3.1.3 | Board-level power delivery requirements are documented: all VRM specifications, rail voltages and tolerances, power sequencing requirements, and load transient specifications are provided to the board design team. | <input type="checkbox"/> Pending |       |                  |
| 3.1.4 | Cooling solution requirements are documented: TDP, airflow direction, minimum airflow velocity, heatsink specification, thermal interface material (TIM) spec, and mechanical mounting requirements.                  | <input type="checkbox"/> Pending |       |                  |
| 3.1.5 | Mechanical form factor is defined and documented (FHFL PCIe, OAM, custom module, or other); dimensions,                                                                                                               | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                               | Status                           | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | connector placement, keep-out zones, and mounting hole locations are specified.                                                                                                                              |                                  |       |                  |
| 3.1.6 | All debug interfaces are defined and specified: JTAG (IEEE 1149.1), SWD, UART/UART-over-USB, and internal scan chain access; debug port physical protection (lockdown in production) is specified.           | <input type="checkbox"/> Pending |       |                  |
| 3.1.7 | Physical layer (PHY) for all high-speed interfaces has been verified by simulation and/or characterization ; equalization settings, pre-emphasis, and receiver sensitivity margins are within specification. | <input type="checkbox"/> Pending |       |                  |
| 3.1.8 | Signal integrity (SI) analysis is complete for all high-speed links including SerDes interfaces and HBM (if present); eye diagram margin, jitter budget, and channel insertion loss are within               | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                                    | Status                           | Owner | Notes / Evidence |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | specification at target data rate.                                                                                                                                                                |                                  |       |                  |
| <b>3.1.9</b>  | EMC/EMI compliance requirements for target markets are identified; design features addressing radiated and conducted emissions are documented; pre-compliance assessment plan is in place.        | <input type="checkbox"/> Pending |       |                  |
| <b>3.1.10</b> | ESD protection is verified on all external-facing interfaces; protection level (IEC 61000-4-2 class) is specified per interface type; protection circuit implementation is reviewed and accepted. | <input type="checkbox"/> Pending |       |                  |

## 3.2 Software & SDK Integration

| #            | Checklist Item                                                                                                                                         | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>3.2.1</b> | Hardware Abstraction Layer (HAL) API is fully documented; all public API functions are specified with parameter types, return values, error codes, and | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                           | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | thread-safety guarantees.                                                                                                                                                                                |                                  |       |                  |
| 3.2.2 | OS driver interface specification (Linux kernel module interface, Windows KMDF/WDF model) is complete and stable; driver ABI is versioned and backward compatibility policy is defined.                  | <input type="checkbox"/> Pending |       |                  |
| 3.2.3 | ML framework compatibility layer (ONNX, TensorFlow, PyTorch) is specified; operator coverage matrix is documented; unsupported operators have a documented fallback path or are explicitly out of scope. | <input type="checkbox"/> Pending |       |                  |
| 3.2.4 | Graph query API is defined; supported query language(s) (Gremlin, openCypher, and/or custom API) are specified with coverage limitations and performance characteristics documented per query type.      | <input type="checkbox"/> Pending |       |                  |
| 3.2.5 | SDK installation and dependency requirements                                                                                                                                                             | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                      | Status                           | Owner | Notes / Evidence |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | are documented (OS versions, runtime dependencies, compiler versions, third-party libraries); installation has been validated on all supported host platforms.                                                      |                                  |       |                  |
| 3.2.6 | Programming model guide is complete covering: memory management model, data flow programming paradigm, buffer ownership rules, synchronization primitives, and error handling patterns.                             | <input type="checkbox"/> Pending |       |                  |
| 3.2.7 | Software simulator or hardware emulator is available for pre-silicon software development; simulator functional coverage and known limitations are documented; simulator/hardware divergence test plan is in place. | <input type="checkbox"/> Pending |       |                  |
| 3.2.8 | Debugging and profiling tools are specified and available: performance counter definitions, hardware event                                                                                                          | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | trace interface, profiler data format, and integration with at least one standard profiling framework (e.g., perf, VTune) documented.                                                                                 |                                  |       |                  |
| <b>3.2.9</b>  | SDK versioning scheme is defined; backward and forward compatibility policy is documented; API deprecation process is defined with minimum deprecation notice period specified.                                       | <input type="checkbox"/> Pending |       |                  |
| <b>3.2.10</b> | SDK license terms and distribution model are defined; third-party component licenses are identified and compatible with the intended distribution model; open-source component list is current and reviewed by legal. | <input type="checkbox"/> Pending |       |                  |

### 3.3 System-Level Integration

| #     | Checklist Item                                                                                                                                                                                       | Status                           | Owner | Notes / Evidence |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 3.3.1 | Reference platform integration guide is available; the guide covers board bring-up sequence, firmware loading, software stack installation, and validation test execution on the reference platform. | <input type="checkbox"/> Pending |       |                  |
| 3.3.2 | Boot ROM and firmware signing chain is fully defined; all signing keys are generated, stored under HSM protection, and the chain of trust from Boot ROM through all firmware stages is documented.   | <input type="checkbox"/> Pending |       |                  |
| 3.3.3 | Secure Boot is implemented and verified end-to-end; tampered firmware is rejected with correct error reporting; rollback protection is implemented; verification results are documented.             | <input type="checkbox"/> Pending |       |                  |
| 3.3.4 | Firmware update mechanism (OTA and/or USB) is documented, including update package format,                                                                                                           | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                           | Status                           | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | authentication, integrity verification, A/B partition scheme (if used), rollback behavior, and failure recovery.                                                                                                         |                                  |       |                  |
| 3.3.5 | System-level power management states (S0 active through S5 mechanical off, or equivalent for the target platform) are supported; wake-on-event capability requirements are defined and implemented.                      | <input type="checkbox"/> Pending |       |                  |
| 3.3.6 | OS driver compatibility is verified for all supported operating systems; minimum Linux kernel version and Windows driver model version are specified and tested; driver WHQL submission plan (if applicable) is defined. | <input type="checkbox"/> Pending |       |                  |
| 3.3.7 | Hypervisor and virtualization support is defined; SR-IOV implementation (if applicable) is specified and verified; virtual function isolation, guest driver model, and live                                              | <input type="checkbox"/> Pending |       |                  |

| #             | Checklist Item                                                                                                                                                                                                     | Status                           | Owner | Notes / Evidence |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|               | migration behavior (if applicable) are documented.                                                                                                                                                                 |                                  |       |                  |
| <b>3.3.8</b>  | Multi-chip interconnect (if applicable) is specified and verified: inter-chip link protocol, bandwidth, latency, error handling, and synchronization are all documented and simulation-validated.                  | <input type="checkbox"/> Pending |       |                  |
| <b>3.3.9</b>  | Inter-chip synchronization protocol is defined and latency-bounded; clock distribution architecture for multi-chip configurations is specified; synchronization verification approach is documented.               | <input type="checkbox"/> Pending |       |                  |
| <b>3.3.10</b> | System reset and recovery behavior is defined for all reset types (power-on reset, warm reset, software-initiated reset, watchdog-triggered reset); recovery time from each reset type is specified and validated. | <input type="checkbox"/> Pending |       |                  |

## SECTION 4: RISK ASSESSMENT

This section validates the completeness and adequacy of risk identification, analysis, and mitigation planning for all risk categories applicable to the MGLC-XA program. For each item, the Status field indicates whether the risk has been adequately assessed and mitigated to an accepted residual risk level — not whether the risk itself has been eliminated.

### Note on Risk Status Codes

In this section,

**Pass**

means the risk is documented, assessed, and mitigated to an accepted residual level with an owner assigned.

**Partial**

means the risk is identified but mitigation planning is incomplete.

**Fail**

means the risk has not been adequately assessed or presents an unacceptable unmitigated residual risk to the program.

### 4.1 Technical Risks

| #     | Checklist Item                                                                      | Status                           | Owner | Notes / Evidence |
|-------|-------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.1.1 | All architecture-level risks are documented in a formal risk register with risk ID, | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | description, probability, impact, risk score, mitigation actions, residual risk, and owner for each entry.                                                                                                            |                                  |       |                  |
| 4.1.2 | Single Points of Failure (SPOFs) in the architecture are identified and each has either a redundancy mechanism, a graceful degradation mode, or a documented acceptance rationale approved by the architecture board. | <input type="checkbox"/> Pending |       |                  |
| 4.1.3 | Data corruption risk scenarios are enumerated for all data paths; countermeasures (ECC, CRC, end-to-end data integrity checks) are defined and implemented for each scenario; residual risk is accepted.              | <input type="checkbox"/> Pending |       |                  |
| 4.1.4 | Deadlock risk analysis is complete for all state machines, queues, and inter-module dependencies; all identified potential deadlock conditions are resolved by                                                        | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | design or formally accepted with detection/recovery mechanisms.                                                                                                                                                             |                                  |       |                  |
| 4.1.5 | Race condition analysis is complete for all shared resources (memories, registers, semaphores, queues); all identified race conditions are resolved by design; resolution is verified through simulation or formal methods. | <input type="checkbox"/> Pending |       |                  |
| 4.1.6 | Timing closure risk for the target process node is assessed; preliminary static timing analysis results are reviewed; high-risk paths are identified and assigned to engineering teams for resolution.                      | <input type="checkbox"/> Pending |       |                  |
| 4.1.7 | Yield risk for the target ASIC process node is assessed; die area, process defect density, and target yield model are reviewed; yield enhancement features (redundancy, repair fuses) are identified if required.           | <input type="checkbox"/> Pending |       |                  |

| #      | Checklist Item                                                                                                                                                                                                         | Status                           | Owner | Notes / Evidence |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.1.8  | Third-party IP (hard macros, licensed soft IP, memory compilers) integration risk is assessed for each IP block; IP qualification status, silicon-proven status, and supplier support terms are documented.            | <input type="checkbox"/> Pending |       |                  |
| 4.1.9  | Foundry process node availability risk is assessed; risk of node discontinuation, capacity shortage, or geopolitical supply disruption is evaluated; alternate process node feasibility is assessed as contingency.    | <input type="checkbox"/> Pending |       |                  |
| 4.1.10 | Post-silicon debug risk is assessed; DFT (Design-for-Test) coverage target of $\geq 95\%$ structural fault coverage is defined; preliminary DFT insertion plan and DFT coverage estimate are reviewed by the DFT lead. | <input type="checkbox"/> Pending |       |                  |

## 4.2 Supply Chain & Manufacturing Risks

| #     | Checklist Item                                                                                                                                                                                                       | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.2.1 | Primary foundry is identified and a business relationship (NDA, PDK access, tape-out scheduling, foundry engineering support) is formally established with signed agreements.                                        | <input type="checkbox"/> Pending |       |                  |
| 4.2.2 | Backup foundry or second-source manufacturing option is assessed; technology transfer feasibility and incremental cost and schedule impact of a backup foundry scenario are documented.                              | <input type="checkbox"/> Pending |       |                  |
| 4.2.3 | Long lead-time component risks are identified for all critical components (HBM memory, SerDes PHY IP, specialized passive components, packaging materials); procurement strategy and buffer stock plans are defined. | <input type="checkbox"/> Pending |       |                  |
| 4.2.4 | Bill of Materials (BOM) cost risk analysis is complete; cost sensitivity to commodity                                                                                                                                | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | pricing fluctuations (memory, substrates) is quantified; cost contingency provisions are reflected in the program budget.                                                                                   |                                  |       |                  |
| 4.2.5 | Packaging technology risk (2.5D interposer, 3D stacking, or advanced fan-out as applicable) is assessed; packaging supplier qualification status and yield history on the selected technology are reviewed. | <input type="checkbox"/> Pending |       |                  |
| 4.2.6 | Test strategy is defined; Automated Test Equipment (ATE) availability and capacity for the production test volume are confirmed; test program development schedule is reviewed and feasible.                | <input type="checkbox"/> Pending |       |                  |
| 4.2.7 | Minimum Order Quantity (MOQ) constraints for all critical components are documented; MOQ exposure relative to demand forecast is quantified;                                                                | <input type="checkbox"/> Pending |       |                  |

| # | Checklist Item                                               | Status | Owner | Notes / Evidence |
|---|--------------------------------------------------------------|--------|-------|------------------|
|   | inventory liability risk is reflected in the financial plan. |        |       |                  |

## 4.3 Security Risks

| #     | Checklist Item                                                                                                                                                                                                                                    | Status                           | Owner | Notes / Evidence |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.3.1 | A formal threat model document exists for the MGLC-XA; the threat model uses a recognized methodology (e.g., STRIDE, PASTA); all threat categories are covered; the document has been reviewed and accepted by the security lead.                 | <input type="checkbox"/> Pending |       |                  |
| 4.3.2 | Side-channel attack risk (power analysis, timing analysis, electromagnetic analysis) is assessed; relevant attack surface for the MGLC-XA target deployment environment is evaluated; countermeasures are implemented where risk is unacceptable. | <input type="checkbox"/> Pending |       |                  |
| 4.3.3 | Secure key storage implementation                                                                                                                                                                                                                 | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                     | Status                           | Owner | Notes / Evidence |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | is reviewed; cryptographic keys used for firmware signing, device identity, and data encryption are stored in hardware-protected key storage; key lifecycle management is documented.                              |                                  |       |                  |
| 4.3.4 | Firmware tamper detection is implemented; unauthorized modification of firmware at rest or in transit is detectable and results in device entry into a safe, non-operational state with appropriate event logging. | <input type="checkbox"/> Pending |       |                  |
| 4.3.5 | Debug interface lockdown in production mode is verified: JTAG, SWD, and all internal scan chain access are disabled or require authenticated activation in production-fused devices; lockdown is irreversible.     | <input type="checkbox"/> Pending |       |                  |
| 4.3.6 | Supply chain integrity controls are defined; anti-counterfeiting measures (device                                                                                                                                  | <input type="checkbox"/> Pending |       |                  |

| #            | Checklist Item                                                                                                                                                                                               | Status                           | Owner | Notes / Evidence |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|              | authentication, unique device ID tied to manufacturing records) are implemented and the verification process is documented.                                                                                  |                                  |       |                  |
| <b>4.3.7</b> | Data exfiltration attack surface for multi-tenant deployments is assessed; hardware-enforced isolation between tenant address spaces, DMA domains, and cryptographic key domains is verified and documented. | <input type="checkbox"/> Pending |       |                  |

## 4.4 Regulatory & Compliance Risks

| #            | Checklist Item                                                                                                                                                                                                                              | Status                           | Owner | Notes / Evidence |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| <b>4.4.1</b> | Export control classification is completed by legal/compliance; ECCN classification (or equivalent national export classification) is documented; any applicable license requirements are identified and plans are in place to obtain them. | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                                   | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.4.2 | FCC (Part 15), CE (Red Directive / EMC Directive), UKCA, and RoHS compliance requirements are identified for all target markets; compliance testing plan is in place with test labs identified and scheduled.                    | <input type="checkbox"/> Pending |       |                  |
| 4.4.3 | JEDEC reliability standard compliance requirements are assessed; required qualification tests (HTOL, ELFR, TC, THB, HAST) are identified per the target application class; test vehicle and qualification schedule are defined.  | <input type="checkbox"/> Pending |       |                  |
| 4.4.4 | Functional safety standard applicability is assessed; if ISO 26262, IEC 61508, or another functional safety standard applies to any target application, ASIL/SIL target level is defined and design implications are documented. | <input type="checkbox"/> Pending |       |                  |
| 4.4.5 | Data privacy compliance is assessed for                                                                                                                                                                                          | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                                    | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | edge deployment use cases; if personal data may be processed, GDPR (or equivalent) implications for data residency, encryption at rest, and deletion are identified and addressed in the design.                                  |                                  |       |                  |
| 4.4.6 | Open-source license compliance audit is complete; all open-source components used in firmware, SDK, and toolchain are identified; license obligations (attribution, copyleft, linking restrictions) are documented and fulfilled. | <input type="checkbox"/> Pending |       |                  |

## 4.5 Business & Execution Risks

| #     | Checklist Item                                                                                                                         | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
| 4.5.1 | Schedule risk to the tape-out date is assessed; key risk factors (IP delivery schedule, EDA tool issues, DFT completion, signoff cycle | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                        | Status                           | Owner | Notes / Evidence |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | time) are identified and mitigation actions with owners are documented in the risk register.                                                                                                                          |                                  |       |                  |
| 4.5.2 | ASIC NRE cost overrun risk is assessed; contingency reserve (recommended: $\geq 15\%$ of NRE estimate) is included in the approved program budget; cost overrun triggers and escalation path are defined.             | <input type="checkbox"/> Pending |       |                  |
| 4.5.3 | Talent and key-person dependencies are identified; for each critical role with a single-person dependency, succession or coverage plan (cross-training, documentation, contractor backup) is defined and in progress. | <input type="checkbox"/> Pending |       |                  |
| 4.5.4 | Customer concentration risk is assessed; design-win pipeline diversification strategy is documented; dependence on any single customer for $>40\%$ of projected first-year revenue is                                 | <input type="checkbox"/> Pending |       |                  |

| #     | Checklist Item                                                                                                                                                                                                                   | Status                           | Owner | Notes / Evidence |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------|------------------|
|       | identified and addressed in the business plan.                                                                                                                                                                                   |                                  |       |                  |
| 4.5.5 | IP infringement risk: Freedom-to-Operate (FTO) analysis is complete or formally in progress; any identified IP conflict has a documented resolution path (design-around, license, or legal opinion); legal counsel has reviewed. | <input type="checkbox"/> Pending |       |                  |
| 4.5.6 | Competitive disruption risk is assessed; key competitive offerings in the accelerator and graph-processing market are identified; differentiation strategy is documented and validated against the competitive landscape.        | <input type="checkbox"/> Pending |       |                  |
| 4.5.7 | Partnership and ecosystem risk is assessed; SDK adoption barriers are identified; developer relations, ISV engagement, and reference application strategy are documented to mitigate ecosystem adoption risk.                    | <input type="checkbox"/> Pending |       |                  |

## SECTION 5: REVIEW SIGN-OFF

&

## APPROVALS

### 5.1 Reviewer Sign-Off Table

Each designated reviewer must sign off on the sections within their domain of responsibility before a gate decision can be rendered. Sign-off confirms that the reviewer has personally reviewed all items within their scope and agrees with the status designations recorded.

| Role                       | Name | Signature | Date | Gate 1<br>Approved | Gate 2<br>Approved | Gate 3<br>Approved | Gate 4<br>Approved |
|----------------------------|------|-----------|------|--------------------|--------------------|--------------------|--------------------|
| Chief Architect            |      |           |      | Y / N              | Y / N              | Y / N              | Y / N              |
| VP Engineering             |      |           |      | Y / N              | Y / N              | Y / N              | Y / N              |
| Hardware Lead              |      |           |      | Y / N              | Y / N              | Y / N              | Y / N              |
| Software Lead              |      |           |      | N/A                | Y / N              | Y / N              | Y / N              |
| Security Lead              |      |           |      | Y / N              | Y / N              | Y / N              | Y / N              |
| Quality / Reliability Lead |      |           |      | N/A                | Y / N              | Y / N              | Y / N              |
| Product Manager            |      |           |      | Y / N              | Y / N              | Y / N              | Y / N              |

| Role                          | Name | Signature | Date | Gate 1 Approved | Gate 2 Approved | Gate 3 Approved | Gate 4 Approved |
|-------------------------------|------|-----------|------|-----------------|-----------------|-----------------|-----------------|
| <b>CFO / Finance Approver</b> |      |           |      | N/A             | N/A             | Y / N           | Y / N           |

## 5.2 Open Issues Log

All Fail, Partial, and escalated Pending items must be logged here with a target resolution date and an assigned owner. This log must be reviewed at every gate review meeting and updated at each milestone.

| Issue ID | Description                                 | Checklist Ref | Severity        | Owner | Target Resolution | Status |
|----------|---------------------------------------------|---------------|-----------------|-------|-------------------|--------|
| ISS-001  | [Issue description — enter when identified] | [e.g., 1.2.5] | <b>Critical</b> |       |                   | Open   |
| ISS-002  |                                             |               | <b>High</b>     |       |                   |        |
| ISS-003  |                                             |               | <b>Medium</b>   |       |                   |        |
| ISS-004  |                                             |               | <b>Low</b>      |       |                   |        |
| ISS-005  |                                             |               |                 |       |                   |        |

Severity Definitions: **Critical** — Blocks gate. Resolution required before gate sign-off. | **High** — Resolution required within 2 weeks of gate review. | **Medium** — Resolution required before next gate. | **Low** — Track to closure; does not block any gate.

## 5.3 Gate Decision Summary

The Review Board Chair records the formal gate decision for each applicable gate. A *Conditional Pass* requires all conditions to be met and verified by the designated verifier before the gate is considered fully passed.

| Gate          | Name               | Decision                         | Condition Notes / Outstanding Actions | Decision Date | Decision Authority |
|---------------|--------------------|----------------------------------|---------------------------------------|---------------|--------------------|
| <b>Gate 1</b> | Design Freeze      | <input type="checkbox"/> Pending |                                       |               |                    |
| <b>Gate 2</b> | FPGA Validation    | <input type="checkbox"/> Pending |                                       |               |                    |
| <b>Gate 3</b> | ASIC Tape-Out      | <input type="checkbox"/> Pending |                                       |               |                    |
| <b>Gate 4</b> | Production Release | <input type="checkbox"/> Pending |                                       |               |                    |

Decision values:  **Pass** — All items pass, no outstanding conditions. |  **Conditional Pass** — Gate passed subject to conditions listed. |  **Fail** — Gate not passed. Program must address all Fail items and re-enter review.

## 5.4 Next Review Date

| Next Scheduled Review | Review Type                                  | Convening Authority       | Distribution List             |
|-----------------------|----------------------------------------------|---------------------------|-------------------------------|
| [Date TBD]            | [Gate Review / Interim Review / Risk Update] | [Review Board Chair Name] | [Team DL / Distribution List] |

## 5.5 Document Control

| Attribute                       | Value                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Document Owner</b>           | [Chief Architect / Architecture Review Board Chair]                                                                             |
| <b>Review Frequency</b>         | At each gate milestone, or quarterly at minimum during active development                                                       |
| <b>Version Control Location</b> | [Document Management System URL / Repository Path]                                                                              |
| <b>Distribution List</b>        | [Architecture Team, Hardware Lead, Software Lead, Security Lead, Q/R Lead, Product Manager, VP Engineering, CFO (for Gate 3/4)] |
| <b>Retention Period</b>         | Minimum 10 years from product end-of-life date, or as required by applicable                                                    |

| Attribute      | Value                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------|
|                | regulatory retention obligations                                                                                  |
| Classification | Internal Technical Use — Not for external distribution without explicit written authorization from document owner |